6) '6' :- $6 = 2 \times 3$ [Product of Prime factors.]
It must be divisible by both 2 & '3'.

7) '7' :- Twice of unit digit is substracted from remaining digited number. This process is continued till to get as much as least number & if this least number is divisible by '7', then the given number is divisible by '7', otherwise not divisible.

Ex:- 3857 | 7
-14
3843 | 3
-6
378 | 8
-16
21.

'21' is exactly divisible by '7' then the given number 3857 also divisible by '7'.

8) '8' :- $8 = 2^3$ → Last '3' digits
Last '3' digits of given number exactly divi by '8'.

9) '9' :- Sum of digits div by '9'.

10) '10' :- Last digit must be '0'.

11) '11' :- Substraction of sum of odd & even places (or) Substraction of sum of even & sum of odd places is either zero or multiples of '11'.

Ex:- 1 2 3 4 5 6 3

(sum of odd Places digits) − (sum of even places digits.)
(1+3+5+3) − (2+4+6) = 12−12 = '0' (or)
(sum of even place digits) − (sum of odd Place digits)
(2+4+6) − (1+3+5+3) = 0. So given NO.S is divby '11'.

↳ **No. of factors:-** To find no. of factors, the given no. is (x) converted into product of Powered Prime factors.

$\therefore x = a^m \cdot b^n \cdot c^o \ldots\ldots$ [By Prime factorization]

where a, b, c .... are Prime numbers

m, n, o, ... are '+ve' Integers.

then

No. of factors = $(m+1)(n+1)(o+1)\ldots\ldots$

↳ No. of ways to convert a given number into product of two individual factors is:-

$= \frac{\text{No. of factors}}{2}$ [If factors are even]

$= \frac{\text{No. of factors} - 1}{2}$ [If factors are odd]

Note:- Factors are odd for perfect squares only.

↳ **Sum of factors:-** when $x = a^m \cdot b^n \cdot c^o \ldots\ldots$

Sum of factors = $\left(\frac{a^{m+1}-1}{a-1}\right)\left(\frac{b^{n+1}-1}{b-1}\right)\left(\frac{c^{o+1}-1}{c-1}\right)\ldots\ldots$

↳ **Product of factors:-**

(i) If factors are even

Product of factors = $(\text{given Number (or) } x)^{\frac{\text{No. of factors}}{2}}$

(ii) If factors are odd

Product of factors = $(\text{given No. (or) } x)^{\frac{\text{No. of factors} - 1}{2}} \cdot \sqrt{x}$

$= x^{\frac{\text{factors} - 1}{2}} \cdot \sqrt{x}$